

Ansatz Expressivity and Symmetry Accounting in a 10- and 12-Qubit LiH Benchmark

A hardware-efficient circuit that reaches 3.4 mHa
by leaving the physical sector

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Abstract

LiH/STO-3G is solved variationally in two active spaces — CASCI(2e,5o) at 10 qubits and the complete STO-3G space, FCI(4e,6o), at 12 qubits — with two ansatz families, and each result is scored against exact diagonalisation of the identical second-quantised Hamiltonian. The molecular integrals are produced by a self-contained McMurchie–Davidson STO-3G engine written for this workflow, so the entire pipeline executes on one machine with no external quantum chemistry package.

UCCSD reaches 1.6×10^{-7} mHa of the CASCI reference at 10 qubits and 0.011 mHa of the FCI reference at 12 qubits, in both cases with particle-number leakage at or below 1.4×10^{-7} and no penalty term doing any work.

The result of interest is negative. A hardware-efficient `real_amplitudes` circuit, taken from four layers to eight, improves its energy from 20.15 mHa to 3.41 mHa above the CASCI reference — an apparent factor of six, at a cost of 95,391 circuit executions. Measuring two additional expectation values shows that the improvement is not physical: at eight layers the converged state carries $\langle \hat{N} \rangle - 2 = 7.1 \times 10^{-5}$ and $\langle \hat{S}^2 \rangle = 2.1 \times 10^{-3}$, which price out at 2.19 mHa against a budget of 0.16 mHa. The symmetry the circuit broke is worth more than the error it has left, so the energy is not a LiH energy and is not comparable to the reference at all. The four-layer run, which is numerically far worse, is by contrast a valid measurement of an ansatz limitation. Reported without the symmetry audit, the ordering of those two runs inverts.

The cost of the instrumentation is two expectation values per converged state.

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1 Introduction

1.1 Motivation

A variational quantum eigensolver returns a number. Whether that number is an approximation to a molecular energy depends on a premise that the optimisation does not check: that the converged state lies in the symmetry sector the reference energy was computed in. The variational principle $\langle \psi | \hat{H} | \psi \rangle \geq E_0$ holds against the global ground state of $\hat{H}$, not against the ground state of the N -electron, spin- S sector that the chemistry is about. A state which has leaked out of that sector can sit below the reference while meaning nothing.

LiH/STO-3G is small enough that this can be examined exactly. The full space is six spatial orbitals holding four electrons; both active spaces used here are diagonalisable to machine precision, so the reference is not a higher-level-of-theory estimate but the exact answer for the operator being measured. Any discrepancy is therefore attributable to the ansatz, the optimiser, or the symmetry of the converged state — and the three can be separated.

1.2 System and active spaces

The geometry is the experimental equilibrium bond length $r_e(\text{LiH}) = 1.5949 \text{ \AA}$ (Huber and Herzberg, via NIST CCCBDB), rounded to $1.595 \text{ \AA}$, in vacuum, closed shell, in the STO-3G basis.

This is byte-identical to the geometry used by the QARP/Qulacs implementation of the same system in `../lih`, and `tests/test_core.py` parses that file and fails if the two ever diverge.

Table 1: The two active spaces. The reference follows the space: with `frozen-core` the target is CASCI(2e,5o), *not* full-space FCI, and an energy below it is evidence that the state left the two-electron sector rather than evidence of better chemistry.

-space	Active space	Frozen	Qubits	Pauli terms	Reference
<code>frozen-core</code>	CASCI(2e,5o)	Li 1s	10	276	CASCI in that space
<code>full</code>	FCI(4e,6o)	—	12	631	FCI in STO-3G

Freezing the lithium 1s pair costs 0.227 mHa of the 20.378 mHa of correlation energy available in the complete space, which is why the smaller problem is the better-posed one: it removes two qubits and 355 Pauli terms at a cost roughly one seventh of chemical accuracy.

1.3 Scope and claims

This report makes four claims and declines several others; §5 lists the declined ones.

1. UCCSD attains chemical accuracy against exact diagonalisation in both active spaces, and does so with particle-number leakage that is structural rather than penalised.
2. A hardware-efficient `real_amplitudes` circuit does not, at either four or eight layers.
3. The eight-layer improvement over the four-layer run is an artefact of symmetry breaking, and is identified as such by two expectation values that cost nothing to measure relative to the optimisation itself.
4. The integral engine written for this workflow reproduces published LiH/STO-3G RHF and FCI energies, and the second-quantised Hamiltonian it produces reproduces the SCF energy on the Hartree–Fock determinant to ten decimal places in both active spaces.

1.4 Principal results

Headline

UCCSD reaches 1.6×10^{-7} mHa of CASCI at 10 qubits in 48 seconds and 641 circuit executions. The eight-layer hardware-efficient run reaches 3.41 mHa in 1,473 seconds and 95,391 circuit executions, and that number is not usable: its symmetry contamination prices at 2.19 mHa, which exceeds the error it claims to have left.

2 Methodology

2.1 Integral construction

PySCF publishes no Windows wheel and its source build is unsupported there, so `integrals.py` implements STO-3G one- and two-electron integrals by the McMurchie–Davidson scheme, together with closed-shell restricted Hartree–Fock, in NumPy alone. At six basis functions this is a smaller and more auditable dependency than a Fortran-backed chemistry package. `prepare` uses PySCF when it is importable and this engine otherwise; the engine actually used is written into the cache as `integral_backend`, so a result always records its own provenance. Both results reported here carry `integral_backend: native-sto3g`.

The engine is deliberately narrow: STO-3G, closed shell, *s* and *p* functions. It refuses other bases and unknown elements rather than guessing.

2.2 Validation protocol

`tests/test_integrals.py` pins the engine against three independent references:

1. the H₂/STO-3G energy tabulated by Szabo and Ostlund;
2. published LiH/STO-3G RHF and FCI energies;
3. the requirement that the Hartree–Fock determinant of the *second-quantised* Hamiltonian equal the SCF energy to ten decimal places in both active spaces.

The third is the strongest check available without a second chemistry code, because it ties the CAS transformation, the two-electron integral transformation and the frozen-core energy together: an error in any one of them breaks it.

Separately, `run.py validate` diagonalises the *penalised* Hamiltonian exactly and fails if its ground state does not lie in the target electron sector. It writes a receipt bound by SHA-256 to the cache, the molecular specification, the workflow source and the penalty settings. `run.py vqe` refuses to start without a matching receipt, so editing any workflow file invalidates every downstream result rather than silently producing a mismatched one.

2.3 Acceptance criteria

A run is recorded as `verified_chemical_accuracy` only if all three hold:

1. $|E_{\text{VQE}} - E_{\text{ref}}| < 1.6 \text{ mHa}$ for the identical cached Hamiltonian;
2. the symmetry contamination, priced at 1 Ha per unit of $(\langle \hat{N} \rangle - N_{\text{target}})^2$ and per unit of $\langle \hat{S}^2 \rangle$, totals less than 0.16 mHa — one tenth of chemical accuracy, small enough that the contamination cannot be what produced the reported error;
3. where the matrix-product-state simulator was used, its converged energy agrees with an exact statevector re-evaluation to within the same 0.16 mHa.

Criterion (2) is energetic rather than a demand for zero leakage. A hardware-efficient ansatz is not particle-number conserving and always leaks a little; a criterion of exact conservation would fail every physically correct run of that family. Pricing the leak instead makes the comparison decidable.

On the meaning of “chemical accuracy” here

The threshold is 1.6 mHa against the reference *for this Hamiltonian*. STO-3G itself sits roughly 1 kcal/mol from the basis-set limit, so every statement in this report concerns the solver, not experiment.

2.4 Computational details

Qiskit 2.5.2, qiskit-aer 0.17.2, qiskit-nature 0.8.0, OpenFermion 1.8.1, NumPy 2.5.2, SciPy 1.18.0, on Python 3.12.13 under Windows 11. Optimiser L-BFGS-B, tolerance 10^{-10} , maximum 500 iterations, seed 7. Simulator method `statevector` for every run reported here.

UCCSD gradients are central finite differences. Parameter shift is not applicable: each UCCSD parameter drives several rotations within one excitation generator, so the two-term shift rule does not hold, and the runtime detects this and refuses rather than returning a silently wrong gradient. The `real_amplitudes` runs do use parameter shift, for which the precondition does hold.

3 Results

3.1 Reference energies

Table 2: Exact quantities for the two cached Hamiltonians, from diagonalisation in the N -electron sector.

	Frozen core, 10 qubits	Full space, 12 qubits
$E(\text{RHF})$	$-7.862023874015 \text{ Ha}$	$-7.862023874015 \text{ Ha}$
$E(\text{reference})$	$-7.882174519890 \text{ Ha}$	$-7.882401946215 \text{ Ha}$
Reference method	CASCI(2e,5o)	FCI(4e,6o)
Correlation energy	20.151 mHa	20.378 mHa
Nuclear repulsion	0.995317638094 Ha	0.995317638094 Ha
Core energy	$-6.802973547530 \text{ Ha}$	0.995317638094 Ha
Pauli terms	276	631
Discarded terms	0	0

No Hamiltonian terms were discarded at the 10^{-6} Ha cutoff in either space, so the truncation L_1 bound is exactly zero and contributes nothing to the error budget.

3.2 Variational results

Table 3: The four runs. “RA- n ” denotes `real_amplitudes` with n layers and linear entanglement. Errors are against the reference of Table 2 for the same space.

Ansatz	Space	Params	Error (mHa)	Circuits	Time (s)	Status
UCCSD	10q	24	1.59×10^{-7}	641	47.5	verified
UCCSD	12q	92	0.011001	4,259	2,531.7	verified
RA-4	10q	50	20.150496	$\sim 41,000$	408.8	outside accuracy
RA-8	10q	90	3.407633	95,391	1,473.5	symmetry broken

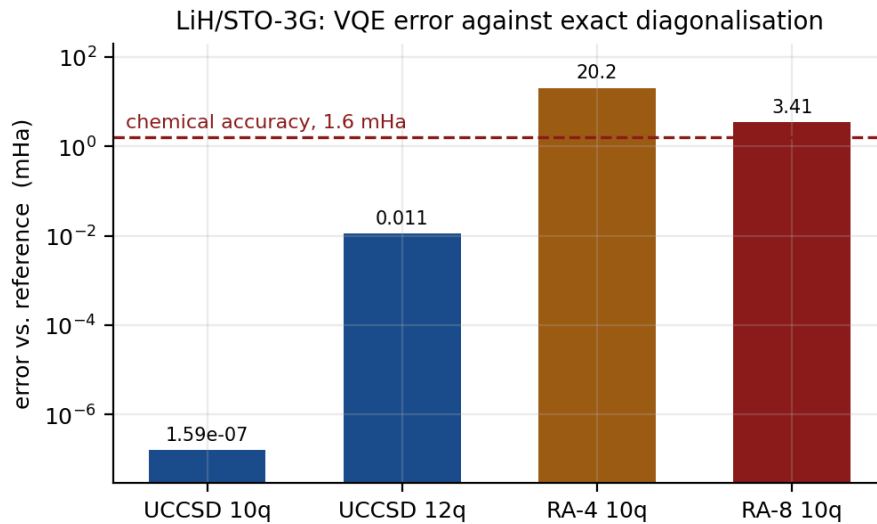


Figure 1: Error against exact diagonalisation, logarithmic. The two UCCSD runs clear chemical accuracy by six orders of magnitude and by a factor of 145 respectively. Read alone, this figure says the eight-layer hardware-efficient run is the better of the two `real_amplitudes` results; Figure 2 says otherwise.

UCCSD at 10 qubits converges in ten L-BFGS-B iterations and twelve objective evaluations. The residual $1.59 \times 10^{-7} \text{ mHa}$ is not an ansatz limitation: CASCI(2e,5o) with two electrons is within the reach of singles and doubles exactly, and what remains is optimiser tolerance. At 12 qubits

the residual rises to 0.011 mHa, still a factor of 145 inside the threshold, and the floor there is finite-difference gradient noise rather than expressivity.

3.3 The symmetry audit

Table 4: Symmetry diagnostics at the converged state. *Priced* is the contamination energy at 1 Ha per unit, against the 0.16 mHa budget of §2.3.

Ansatz	Space	$\langle \hat{N} \rangle - N$	$\langle \hat{S}^2 \rangle$	Priced (mHa)	Verdict
UCCSD	10q	1.0×10^{-13}	2.60×10^{-9}	0.0000026	inside
UCCSD	12q	8.4×10^{-13}	1.37×10^{-7}	0.000137	inside
RA-4	10q	3.2×10^{-8}	3.46×10^{-10}	0.000064	inside
RA-8	10q	7.1×10^{-5}	2.05×10^{-3}	2.191	outside

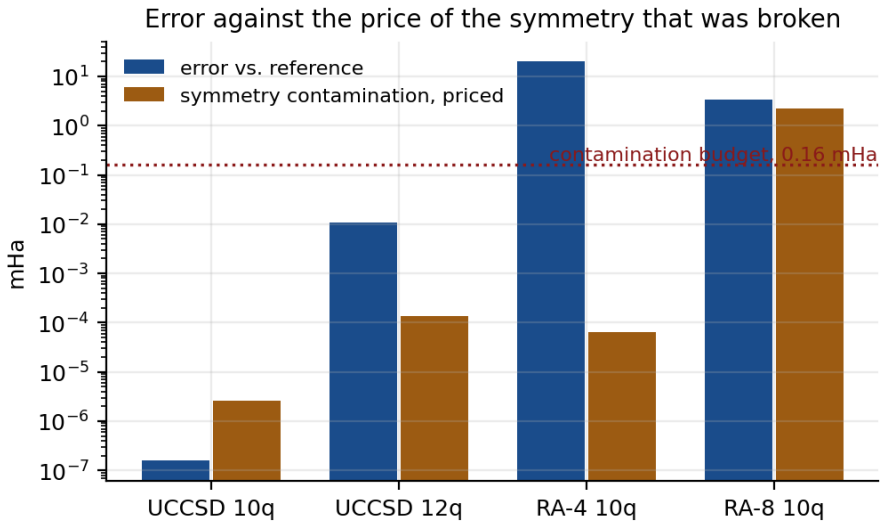


Figure 2: Error beside the price of the symmetry that was broken to obtain it. For the first three runs the contamination is orders of magnitude below the error, and the error is therefore a statement about the ansatz. For RA-8 the contamination exceeds the error.

This is the central observation of the report. Doubling the depth of the hardware-efficient circuit from four layers to eight moves the energy from 20.15 mHa to 3.41 mHa above the reference. Presented as Table 3 alone, that is a sixfold improvement and reads as progress toward chemical accuracy: another doubling would appear to be all that is required.

Table 4 shows what actually happened. The eight-layer circuit bought its energy by leaving the two-electron singlet sector. Its particle-number deviation is three orders of magnitude larger than the four-layer run’s, its $\langle \hat{S}^2 \rangle$ is seven orders larger, and together these price at 2.191 mHa — 13.7 times the budget, and 0.79 mHa *more* than the error it claims to have left. There is no residual against which to compare a further improvement, because the quantity being reported is not an energy of LiH in the target sector.

The four-layer run, which by the energy alone looks like the failure, is the scientifically usable measurement of the two: its contamination is 6.4×10^{-5} mHa, so its 20.15 mHa error is an honest statement that a four-layer linear-entanglement `real_amplitudes` circuit recovers essentially none of the correlation energy of this system. The audit inverts the ordering of the two runs.

What Table 3 would say without Table 4

“Increasing `real_amplitudes` depth from four to eight layers reduces the error against CASCI by a factor of 5.9, to 3.41 mHa, suggesting that chemical accuracy is reachable with a hardware-efficient ansatz at modest additional depth.” Every number in that sentence is correct. The conclusion does not follow.

3.4 Cost

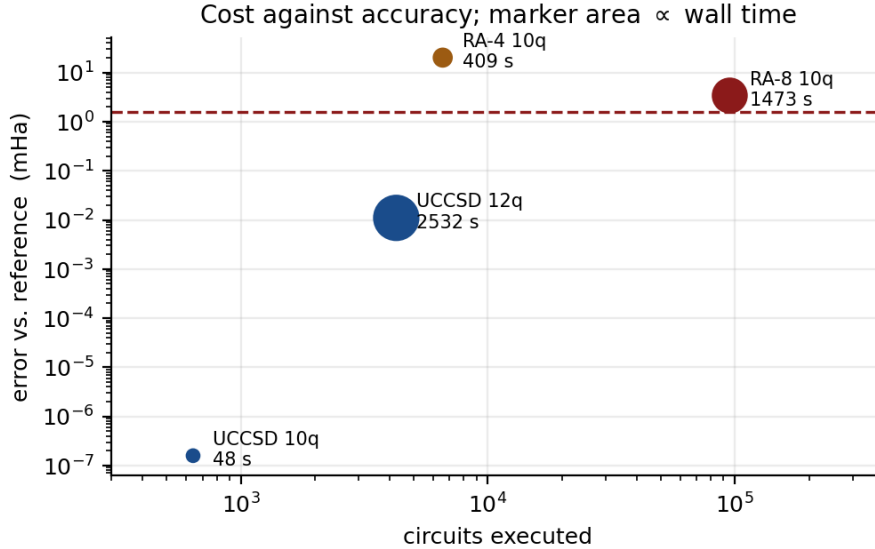


Figure 3: Circuit executions against error; marker area proportional to wall time. The two families are separated by roughly two orders of magnitude in cost at every accuracy.

The cost asymmetry is as large as the accuracy asymmetry and has the same cause. UCCSD converges in 10 and 18 optimiser iterations respectively because its doubles amplitudes have a finite gradient at the Hartree–Fock reference — the same quantity that makes MP2 non-zero. The hardware-efficient circuit does not: at $\theta = 0$ it *is* the Hartree–Fock determinant, and because the Hamiltonian conserves particle number, a lone R_y excitation out of that determinant has exactly zero gradient. Measured live-parameter counts at $\theta = 0$ are 0/50 in the frozen-core space and 0/60 in the full space; L-BFGS-B started there returns after zero iterations reporting clean convergence at E_{HF} .

The workaround is a Gaussian kick, and its magnitude is not a free parameter. Measured on the self-test’s synthetic two-electron 10-qubit system with four layers:

Table 5: Initialisation study, `real_amplitudes`, four layers, L-BFGS-B to convergence.

Start	Result
$\theta = 0$ exactly	0 iterations, stuck at E_{HF}
$\theta = 0 + N(0, 0.05)$	converges, 0.0% of correlation recovered
$\theta = 0 + N(0, 0.2)$	converges, 0.0% recovered
$\theta = 0 + N(0, 0.5)$	converges 2.20 Ha <i>above</i> E_{HF}
uniform $[-\pi, \pi]$	fails to converge, 1.29 Ha above E_{HF}

Near $\theta = 0$ the landscape is flat; further out it is populated with high local minima. A larger kick is not a better one. On real LiH the default $\sigma = 0.05$ does let the optimiser move, and it still recovers essentially none of the correlation — so the limitation is the ansatz, not the start, and that conclusion required distinguishing the two.

3.5 Simulator method

`matrix_product_state` is the package default because it is the method that scales to the larger active spaces this workflow is a warm-up for, and because `vqe` re-evaluates its converged energy with `statevector` and refuses to claim accuracy if the two disagree. At *this* size it is not the fast choice. Measured on this machine at 10 qubits with an ~ 880 -term objective, one L-BFGS-B iteration of 101 circuits over 50 parameters:

Method	Per iteration
<code>matrix_product_state</code>	~ 13 s
<code>statevector</code>	~ 2.4 s

MPS pays a per-Pauli-term contraction cost that a 10-qubit statevector does not have, and UCCSD’s much deeper circuits widen the gap. Every run in Table 3 therefore used `statevector`, and `mps_vs_statevector_hartree` is recorded as exactly 0.0 in each result file.

4 Discussion

4.1 Why the audit is cheap and the failure is not

Measuring $\langle \hat{N} \rangle$ and $\langle \hat{S}^2 \rangle$ at the converged state costs two expectation values against a run that already executed between 641 and 95,391 circuits. In the worst case here the instrumentation is a 2×10^{-5} fraction of the computation it validates. The failure it catches, by contrast, is silent: the optimiser converged cleanly, the energy moved in the right direction, and the number produced lies in a plausible range. Nothing in the optimisation reports a problem, because from the optimiser’s point of view there is none — it minimised the objective it was given, over the whole Hilbert space, and the objective’s lower bound over that space is below the sector ground state.

4.2 Why UCCSD does not need the apparatus

UCCSD is built from fermionic excitation operators with balanced creation and annihilation indices, so it commutes with $\hat{N}$ and $\hat{S}_z$ exactly. The number penalty contributes identically zero, and the measured leak at the optimum is -1.3×10^{-15} in the frozen-core space. The penalty machinery is still applied and still audited; it simply has nothing to correct.

This is worth stating precisely, because the guarantee is narrower than it is often taken to be. UCCSD conserves N and S_z but *not* S^2 at arbitrary amplitudes, since its $\alpha \rightarrow \alpha$ and $\beta \rightarrow \beta$ amplitudes are independent. It becomes a spin eigenstate at the variational minimum, which is where $\langle \hat{S}^2 \rangle$ is measured — and Table 4 records 2.6×10^{-9} and 1.4×10^{-7} , consistent with that expectation but not guaranteed by construction.

4.3 What the self-test does and does not establish

`run.py selftest` proves the stack: operator conversion checked both spectrally and determinant by determinant (the latter is what pins down qubit ordering, since a bit permutation is a similarity transform and leaves the spectrum unchanged), the Hartree–Fock frame rotation, MPS against exact statevector, parameter shift against finite differences, and the requirement that the VQE respect the variational bound, terminate cleanly and never return a point worse than its own start.

It does *not* assert that the VQE reaches the exact answer or beats Hartree–Fock. Its Hamiltonian is random, and per Table 5 the hardware-efficient ansatz genuinely cannot solve it at this filling. A self-test demanding otherwise would be asserting expressivity it cannot promise. Correlation

recovery is printed there as a diagnostic and is not scored. For the claim that the chemistry is right, the evidence is `tests/test_integrals.py` and `run.py validate`.

5 Claims not supported by these data

1. **That `real_amplitudes` cannot reach chemical accuracy on LiH at any depth.** Two depths were run. The eight-layer result is uninterpretable for the reason given above, so the evidence is a single valid point. Deciding this needs a depth scan with the number penalty raised to a validated value at each depth, and $\langle \hat{S}^2 \rangle$ recorded throughout.
2. **That the eight-layer circuit would reach chemical accuracy if the penalty were raised.** Plausible and untested. Raising `-number-penalty` requires re-running `validate` at the new value, and the run was not repeated.
3. **That 3.41 mHa is an upper bound on what the eight-layer circuit can achieve.** The run stopped on TOTAL NO. OF ITERATIONS REACHED LIMIT at 500 iterations, not on convergence. It was still descending.
4. **Anything about hardware.** Every number here is a noiseless simulation. No shot noise, no readout error, no decoherence, no transpilation to a physical coupling map.
5. **That the STO-3G numbers approximate experiment.** They approximate exact diagonalisation in STO-3G, which is roughly 1 kcal/mol from the basis-set limit for LiH.
6. **A comparison against the QARP/Qulacs implementation in `../lih`.** The geometry and basis are pinned equal by a test, which is what would make such a comparison meaningful, but the runs were not executed under matched optimiser settings and are not compared here.

6 Limitations

1. One geometry. No potential-energy curve, so no dissociation energy and no test of whether the conclusions survive into a multireference regime — which for LiH they would not be expected to at 1.595 Å.
2. One basis. The integral engine is STO-3G only by construction.
3. Seed 7 throughout; no statistics over initialisations for the hardware-efficient runs, whose outcome is initialisation-dependent by Table 5.
4. The 1 Ha per unit contamination price is a convention. It is stated so that the budget can be recomputed under a different one, not derived.
5. The RA-4 circuit count is recorded as approximate in the source table; the exact figure is not in the result file, which stores `aer_batches: 66` and the batch composition rather than a total.

7 Conclusions

1. UCCSD solves LiH/STO-3G to 1.6×10^{-7} mHa at 10 qubits and 0.011 mHa at 12 qubits against exact diagonalisation of the identical Hamiltonian, with structurally conserved particle number.
2. The molecular integrals were produced without an external quantum chemistry package, by an engine pinned against published RHF and FCI energies and against the requirement that the second-quantised Hamiltonian reproduce the SCF energy on the reference determinant to ten decimals.

3. A hardware-efficient circuit at eight layers reaches 3.41 mHa and the number is not usable. Its symmetry contamination prices at 2.19 mHa against a 0.16 mHa budget — more than the error it has left. The four-layer run, worse by a factor of six, is the valid measurement of the two.
4. The instrumentation that decides this is two expectation values on the converged state. Against a run of 95,391 circuits, that is not a cost. Without it, Table 3 supports a conclusion that Table 4 refutes.

A Provenance digests

Every result file records the SHA-256 of the cache it read, the molecular specification, and the workflow source that produced it. A mismatch is a hard error, not a warning.

Quantity	Digest (first 16 hex)
Frozen-core specification	2c2041c7740e08d7
Frozen-core cache	6423e19956432afc
Full-space specification	e48966a0f0827801
Full-space cache	b6e5122d87dd5be9
Chemistry workflow	b5eed0d532b10fcb
Validation workflow	04b2e651dd029c5b
Runtime workflow (RA-8, UCCSD-12q)	da1c3cda75f59681

B Complete settings

Setting	Value
Optimiser	L-BFGS-B
Tolerance	10^{-10}
Maximum iterations	500
Seed	7
Simulator method	statevector
MPS truncation threshold	10^{-16} (unused)
Hamiltonian cutoff	10^{-6} Ha
Number penalty	1.0 Ha
Spin penalty	0.0 Ha
Contamination price	1.0 Ha per unit
Contamination budget	1.6×10^{-4} Ha
UCCSD gradient	central finite difference
real_amplitudes gradient	parameter shift
real_amplitudes start	$\theta = 0 + N(0, 0.05)$
UCCSD start	$\theta = 0$ exactly

C Software environment

Python 3.12.13 on Windows 11. qiskit 2.5.2, qiskit-aer 0.17.2, qiskit-nature 0.8.0, OpenFermion 1.8.1, NumPy 2.5.2, SciPy 1.18.0. PySCF absent; integrals from the native engine.

D Reproduction

```
uv venv --python 3.12 .venv && .venv\Scripts\activate
uv pip install -r requirements.txt
```

```
python run.py selftest
python run.py prepare
python run.py validate
python run.py vqe --ansatz uccsd --method statevector --start-noise 0

python run.py prepare --space full
python run.py validate --space full
python run.py vqe --space full --ansatz uccsd \
    --method statevector --start-noise 0

python run.py vqe --ansatz real-amplitudes --layers 4 --method statevector
python run.py vqe --ansatz real-amplitudes --layers 8 --method statevector

python -m unittest discover -s tests
```

E Data inventory

File	Contents
lih_frozen_core_hamiltonian.json	CASCI(2e,5o) cache, 276 Pauli terms
lih_frozen_core_hamiltonian.json.validated.json	its receipt
lih_full_hamiltonian.json	FCI(4e,6o) cache, 631 Pauli terms
lih_full_hamiltonian.json.validated.json	its receipt
lih_frozen_core_vqe_result.json	UCCSD, 10 qubits
lih_full_vqe_uccsd.json	UCCSD, 12 qubits
lih_frozen_core_vqe_he4.json	real_amplitudes, 4 layers
lih_frozen_core_vqe_he8.json	real_amplitudes, 8 layers

All eight files live beside `run.py` and are committed, so every number in this report is recomputable from the repository without re-running anything.

References

1. K. P. Huber and G. Herzberg, *Molecular Spectra and Molecular Structure IV: Constants of Diatomic Molecules*, via NIST CCCBDB.
2. A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry*, Dover, 1996 — $\text{H}_2/\text{STO-3G}$ reference energies.
3. L. E. McMurchie and E. R. Davidson, *J. Comput. Phys.* **26**, 218 (1978) — the integral recursion used by `integrals.py`.